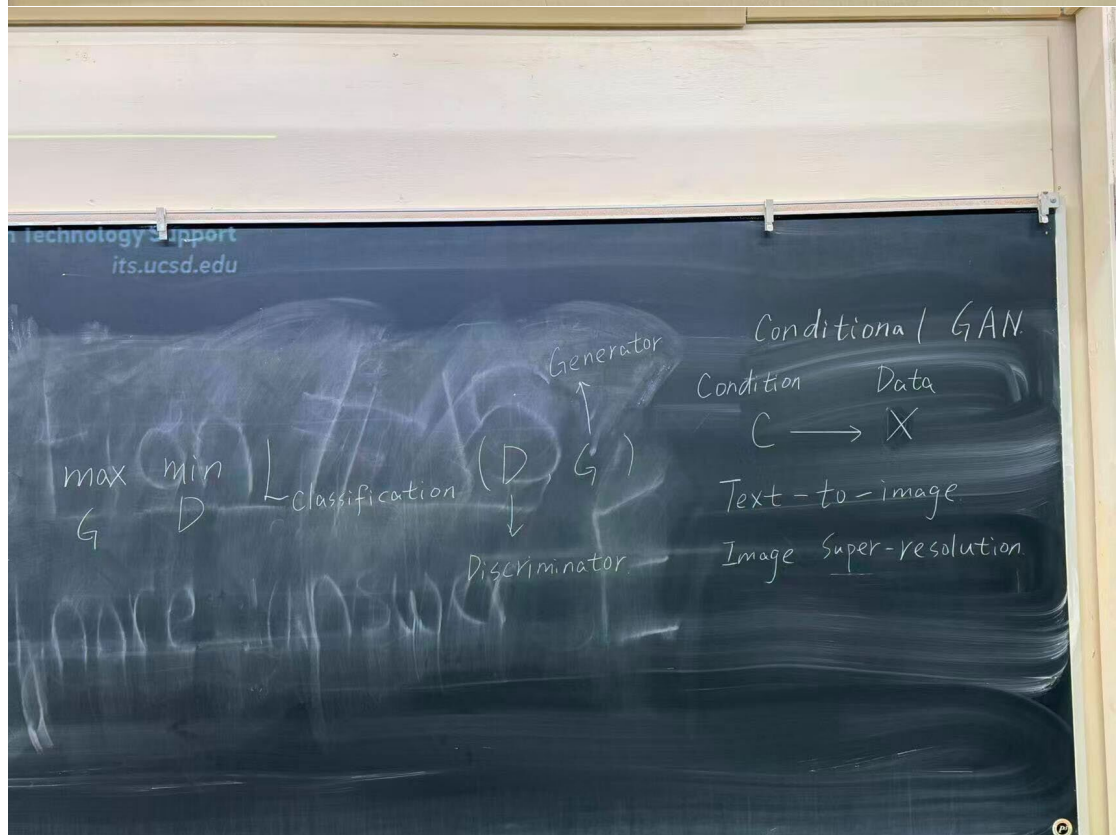
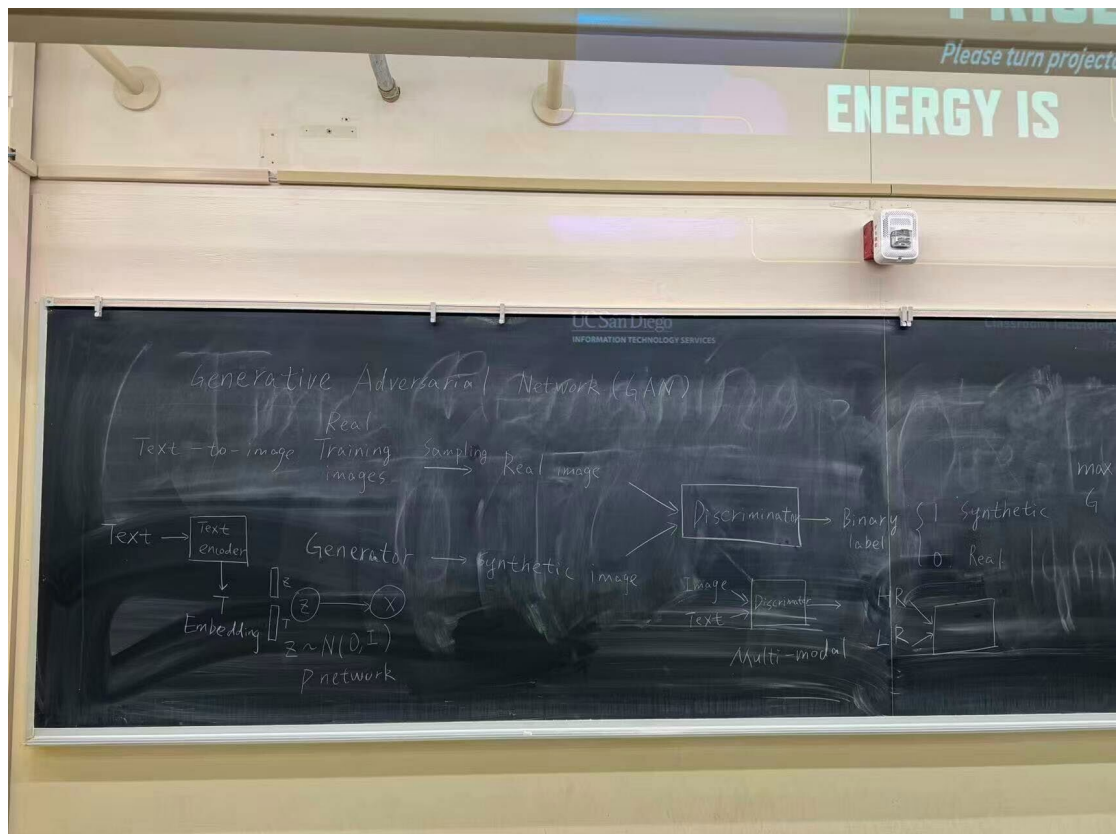




Large language models (LLMs)

- LLM {
- Transformer
 - Training
 - Sampling
 - Prompting
 - Supervised-finetuning (instruction tuning)
 - Parameter-efficient finetuning
 - Reinforcement learning from human feedback (RLHF)
 - Reasoning

Please turn projector off at the end of class.

ENERGY IS

Bayesian network

$$P(W_1, W_2, W_3, \dots, W_n) = P(W_1) \prod_{i=2}^n P(W_i | W_{1:i-1})$$

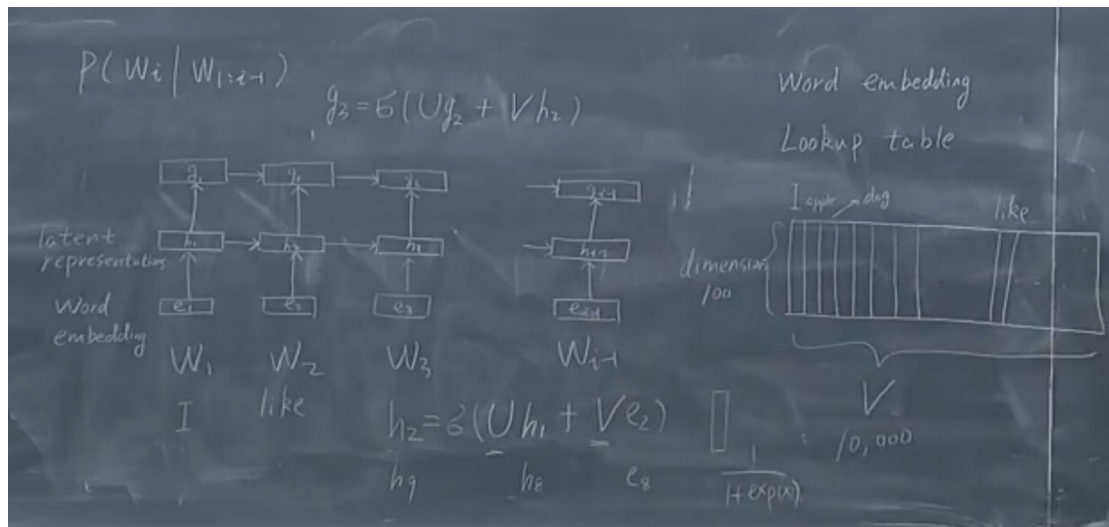


Transformer

Recurrent neural network (RNN) {

- LSTM
- GRU

Sweet
Smelly X
handsome X
sweet ✓



V words in the dictionary.

$10,000$

$P = \text{Softmax}(g_{i-1})$

$P(w_i | w_{1:i-1})$

g_1

g_2

g_3

g_{i-1}

Softmax

P_1

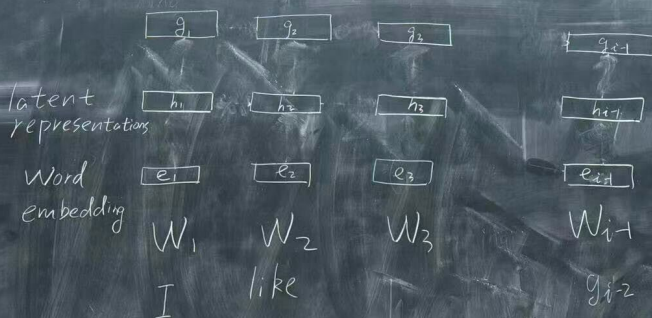
P_2

P_3

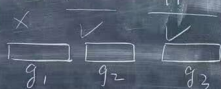
$P_2 = \frac{\exp(g_2)}{\sum_{i=1}^3 \exp(g_i)}$

$P_3 = \frac{\exp(g_3)}{\sum_{i=1}^3 \exp(g_i)}$

$$P(W_i | W_{1:i-1})$$



I like apple



Self-Attention

$$A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{i-2} \end{bmatrix} = \begin{bmatrix} g_1^T g_{i-1} \\ g_2^T g_{i-1} \\ \vdots \\ g_{i-2}^T g_{i-1} \end{bmatrix}$$

because it is



$\hat{A}$

$$\hat{A} = \text{softmax}(A)$$

Smelly X
handsome X
sweet V

10000 next
100

$$g_i = \sum_{j=1}^{i-2} \hat{a}_j g_j$$

$\hat{a}_3, w_3$
 $\hat{a}_1, w_1$
 g_3
 g_1

$$P(s) = P(w_1) \prod_{i=2}^n P(w_i | w_{1:i-1})$$
 Transformer Θ

Training
 Maximum data likelihood

$$\max_{\Theta} \log \prod_{n=1}^N P(s_n)$$

$$h_i = \sum_{j=1}^{i-1} \hat{a}_{ij} \hat{h}_j$$

$$p_i = \text{softmax}(h_i)$$

$$P(w_i | w_{1:i-1})$$

Generate new sentences
 Sampling

$$P(w_{\text{next}} | \text{Current words})$$
 Categorical distribution
 Greedy algorithm

Beam search.

k	step i	step i+1	step i+2
1	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

